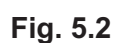


(a) In the source, sucrose is transferred from a mesophyll cell to a phloem sieve tube through a companion cell.

[4]

$$\text{UDPG} + \text{fructose} \xrightleftharpoons{\text{sucrose synthase}} \text{UDP} + \text{sucrose}$$

The structure of UDPG is shown in Fig. 5.2.



..... [1]

(ii) Explain why UDP can be described as a phosphorylated nucleotide.

[2]

(iii) Sucrose synthase acts by using an induced fit mechanism rather than a lock and key mechanism.

With reference to sucrose synthase and the **synthesis** of sucrose, outline the difference between the induced fit mechanism and lock and key mechanism of enzyme action.

[4]

- (c) UDPG is used in some algae (photosynthetic protists) to synthesise a storage compound known as floridean starch.

The molecular structure of floridean starch has been described as an intermediate between amylopectin and glycogen, with little or no amylose.

Describe the molecular structure of floridean starch by completing the passage.

Floridean starch is a polysaccharide composed of monomers.

The monomers are joined by and linkages,

to give a branching structure that is less highly branched than

..... .

[4]

[Total: 15]